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FAST SETTLED AND TRANSIENT LOW DROPOUT (LDO) REGULATOR

Abstract

A voltage regulator includes a pass transistor coupled between an input and an output of the voltage regulator, and an amplifier having a first input configured to receive a reference voltage, and a second input coupled to the output of the voltage regulator via a feedback path. The voltage regulator also includes a voltage booster coupled between an output of the amplifier and a gate of the pass transistor, and a multiplexer having a first input configured to receive a first clock signal, a second input configured to receive a second clock signal having a higher frequency than the first clock signal, and an output coupled to a clock input of the voltage booster. The voltage regulator also includes a detection circuit having an input coupled to the amplifier, and an output coupled to a select input of the multiplexer.

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Background/Summary

BACKGROUND

Field

[0001] Aspects of the present disclosure relate generally to voltage regulators, and more particularly, to low dropout (LDO) regulators.

Background

[0002] Voltage regulators are used in a variety of systems to provide regulated voltages to power circuits in the systems. A commonly used voltage regulator is a low dropout (LDO) regulator. An LDO regulator typically includes a pass transistor and an amplifier coupled in a feedback loop to maintain an approximately constant output voltage based on a reference voltage.

SUMMARY

[0003] The following presents a simplified summary of one or more implementations in order to provide a basic understanding of such implementations. This summary is not an extensive overview of all contemplated implementations and is intended to neither identify key or critical elements of all implementations nor delineate the scope of any or all implementations. Its sole purpose is to present some concepts of one or more implementations in a simplified form as a prelude to the more detailed description that is presented later.

[0004] A first aspect relates to a voltage regulator. The voltage regulator includes a pass transistor coupled between an input of the voltage regulator and an output of the voltage regulator. The voltage regulator also includes an amplifier having a first input, a second input, and an output, wherein the first input of the amplifier is configured to receive a reference voltage, and the second input of the amplifier is coupled to the output of the voltage regulator via a feedback path. The voltage regulator also includes a voltage booster coupled between the output of the amplifier and a gate of the pass transistor, and a multiplexer having a first input, a second input, an output, and a select input, wherein the first input of the multiplexer is configured to receive a first clock signal, the second input of the multiplexer is configured to receive a second clock signal having a higher frequency than the first clock signal, and the output of the multiplexer is coupled to a clock input of the voltage booster. The voltage regulator also includes a detection circuit having an input and an output, wherein the input of the detection circuit is coupled to the amplifier, and the output of the detection circuit is coupled to the select input of the multiplexer.

[0005] A second aspect relates to a method of operating a voltage regulator. The voltage regulator includes a pass transistor, an amplifier, and a voltage booster coupled between an output of the amplifier and a gate of the pass transistor, wherein a first input of the amplifier receives a reference voltage and a second input of the amplifier receives a feedback voltage via a feedback path coupled to a source of the pass transistor. The method includes detecting a difference between the reference voltage and the feedback voltage is greater than a threshold voltage, and switching a clock input of the voltage booster from a first clock signal to a second clock signal in response to detecting the difference between the reference voltage and the feedback voltage is greater than the threshold, wherein the second clock signal has a higher clock frequency than the first clock signal.

[0006] A third aspect relates to a voltage regulator. The voltage regulator includes a pass transistor coupled between an input of the voltage regulator and an output of the voltage regulator, and an amplifier having a first input, a second input, and an output, wherein the first input of the amplifier is configured to receive a reference voltage, and the second input of the amplifier is coupled to the output of the voltage regulator via a feedback path. The voltage regulator also includes a voltage booster coupled between the output of the amplifier and a gate of the pass transistor, wherein the voltage booster includes a boost capacitor and an output capacitor, and the output capacitor is coupled to the gate of the pass transistor. The voltage booster is configured to, during a first phase,

charge the boost capacitor with current from the output of the amplifier, and, during a second phase, boost a voltage on the boost capacitor and transfer charge from the boost capacitor to the output capacitor. The voltage regulator also includes a capacitance controller configured to switch the boost capacitor between a first capacitance and a second capacitance higher than the first capacitance, and a detection circuit having an input and an output, wherein the input of the detection circuit is coupled to the amplifier, and the output of the detection circuit is coupled to the capacitance controller.

[0007] A fourth aspect relates to a method of operating a voltage regulator. The voltage regulator includes a pass transistor, an amplifier, and a voltage booster coupled between an output of the amplifier and a gate of the pass transistor, wherein a first input of the amplifier receives a reference voltage and a second input of the amplifier receives a feedback voltage via a feedback path coupled to a source of the pass transistor. The method includes detecting a difference between the reference voltage and the feedback voltage is greater than a threshold voltage, and switching a boost capacitor of the voltage booster from a first capacitance to a second capacitance in response to detecting the difference between the reference voltage and the feedback voltage is greater than the threshold, wherein the second capacitance is higher than the first capacitance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 shows an example of a low dropout (LDO) regulator.

[0009] FIG. 2 shows an example of an LDO regulator including a voltage divider in a feedback path.

[0010] FIG. 3 shows an example of an LDO regulator including a voltage booster according to certain aspects of the present disclosure.

[0011] FIG. 4 shows an exemplary implementation of the voltage booster according to certain aspects of the present disclosure.

[0012] FIG. 5 shows an exemplary implementation of the voltage booster employing double charge pumping according to certain aspects of the present disclosure.

[0013] FIG. 6 shows an exemplary implementation of switches in the voltage booster of FIG. 5 according to certain aspects of the present disclosure.

[0014] FIG. 7 is a timing diagram illustrating exemplary signals in the voltage booster of FIG. 6 according to certain aspects of the present disclosure.

[0015] FIG. 8 shows an exemplary implementation of a control circuit in the voltage booster of

[0016] FIG. 6 according to certain aspects of the present disclosure.

[0017] FIG. 9 is a timing diagram illustrating exemplary signals in the voltage booster of FIG. 8 according to certain aspects of the present disclosure.

[0018] FIG. 10 shows an example of an LDO regulator including a detection circuit and a multiplexer to reduce settling time according to certain aspects of the present disclosure.

[0019] FIG. 11 shows another example of an LDO regulator including a detection circuit and a multiplexer to reduce settling time according to certain aspects of the present disclosure.

[0020] FIG. 12 shows an exemplary implementation of a detection circuit according to certain aspects of the present disclosure.

[0021] FIG. 13 is a flowchart illustrating an example of a method of operating a voltage regulator according to certain aspects of the present disclosure.

[0022] FIG. 14 shows an example of an LDO regulator including a boost capacitor and a detection circuit configured to switch the boost capacitor between a first capacitance and a second capacitance according certain aspects of the present disclosure.

[0023] FIG. 15 shows another example of an LDO regulator including a boost capacitor and a

detection circuit configured to switch the boost capacitor between a first capacitance and a second capacitance according certain aspects of the present disclosure.

[0024] FIG. **16** shows an example of an LDO regulator including a pulse stretcher according certain aspects of the present disclosure.

[0025] FIG. **17** shows an example implementation of the pulse stretching according to certain aspects of the present disclosure.

[0026] FIG. **18** shows an example of an LDO regulator including a first boost capacitor, a second boost capacitor, and a detection circuit configured to switch the capacitances of the first and second boost capacitors according certain aspects of the present disclosure.

[0027] FIG. **19** shows another example of an LDO regulator including a first boost capacitor, a second boost capacitor, and a detection circuit configured to switch the capacitances of the first and second boost capacitors according certain aspects of the present disclosure.

[0028] FIG. **20** is a flowchart illustrating another example of a method of operating a voltage regulator according to certain aspects of the present disclosure.

DETAILED DESCRIPTION

[0029] The detailed description set forth below, in connection with the appended drawings, is intended as a description of various configurations and is not intended to represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of the various concepts. However, it will be apparent to those skilled in the art that these concepts may be practiced without these specific details. In some instances, well-known structures and components are shown in block diagram form in order to avoid obscuring such concepts.

[0030] FIG. **1** shows an example of a low dropout (LDO) regulator **100** according to certain aspects of the present disclosure. The LDO regulator **100** may be used to provide a noise- sensitive circuit (not shown) with a clean regulated voltage from a noisy supply voltage VCC. The supply voltage VCC may come from a switching regulator (not shown) configured to convert a voltage of a battery into the supply voltage or may come from another voltage source.

[0031] The LDO regulator **100** includes a pass n-type field effect transistor (NFET) **115** and an amplifier **120** (also referred to as an error amplifier). The pass NFET **115** has a drain coupled to the input **105** of the LDO regulator **100**, a gate coupled to the output **126** of the amplifier **120**, and a source coupled to the output **130** of the LDO regulator **100**. The input **105** of the LDO regulator **100** is coupled to a voltage supply rail **112** configured to provide the supply voltage VCC. In FIG. **1**, the current source labeled “I.sub.LOAD” represents the load current of a circuit (not shown) coupled to the output **130** of the LDO regulator **100**.

[0032] The amplifier **120** is configured to control the resistance of the pass NFET **115** between the input **105** and the output **130** of the LDO regulator **100** by adjusting the gate voltage of the pass NFET **115**. For example, the amplifier **120** may increase the resistance of the pass NFET **115** by decreasing the gate voltage of the pass NFET **115**, and decrease the resistance of the pass NFET **115** by increasing the gate voltage of the pass NFET **115**.

[0033] In this example, a reference voltage (denoted “Vref”) is input to a first input **122** (e.g., plus input) of the amplifier **120**. The reference voltage is generated by a reference circuit **160** coupled to the first input **122** of the amplifier **120**. The reference voltage may be implemented with a voltage divider, a bandgap circuit, a circuit including a current source and a resistor coupled in series, or any combination thereof.

[0034] A second input **124** (e.g., minus input) of the amplifier **120** is coupled to the output **130** of the LDO regulator **100** via a feedback path **150**. In this example, the regulated voltage Vreg at the output **130** of the LDO regulator **100** is fed back to the second input **124** of the amplifier **120** via the feedback path **150** to provide a feedback voltage (denoted “Vfb”) at the second input **124** of the amplifier **120**. In the example shown in FIG. **1**, the feedback voltage Vfb is approximately equal to the regulated voltage Vreg. However, it is to be appreciated that the present disclosure is not limited

to this example, as discussed further below.

[0035] During operation, the output **126** of the amplifier **120** drives the gate of the pass NFET **115** in a direction that reduces the difference (i.e., error) between the reference voltage V_{ref} and the feedback voltage V_{fb} . Since the feedback voltage V_{fb} is approximately equal to the regulated voltage V_{reg} in this example, the amplifier **120** drives the gate of the pass NFET **115** in a direction that causes the regulated voltage V_{reg} to be approximately equal to the reference voltage V_{ref} .

[0036] In the example in FIG. **1**, the regulated voltage V_{reg} is fed directly to the second input **124** of the amplifier **120**. However, it is to be appreciated that the present disclosure is not limited to this example. For example, FIG. **2** shows another example of the LDO regulator **100**, in which the regulated voltage V_{ref} is fed back to the amplifier **120** through a voltage divider **215**. The voltage divider **215** includes a first resistor R_1 and a second resistor R_2 coupled in series between the output **130** of the LDO regulator **100** and ground (or some reference potential). In this example, the second input **124** of the amplifier **120** is coupled to a node **220** between the resistors R_1 and R_2 . Thus, in this example, the feedback path **150** between the output **130** of the LDO regulator **100** and the second input **124** of the amplifier **120** includes the voltage divider **215**. In this example, the feedback voltage V_{fb} is related to the regulated voltage V_{reg} as follows:

$$[00001] V_{fb} = \left(\frac{R_2}{R_1 + R_2}\right) \cdot V_{reg} \quad (1)$$

where R_1 and R_2 in equation (1) are the resistances of resistors R_1 and R_2 , respectively. Thus, in this example, the feedback voltage V_{fb} is proportional to the regulated voltage V_{reg} , in which the proportionality is set by the ratio of the resistances of resistors R_1 and R_2 .

[0037] The amplifier **120** drives the gate of the pass NFET **115** in a direction that reduces the difference (i.e., error) between the feedback voltage V_{fb} and reference voltage V_{ref} . This feedback causes the regulated voltage V_{reg} to be approximately equal to:

$$[00002] V_{reg} = \left(1 + \frac{R_1}{R_2}\right) \cdot V_{ref} \quad (2)$$

[0038] As shown in equation (2), in this example, the regulated voltage V_{reg} may be set to a desired voltage by setting the ratio of the resistances of resistors R_1 and R_2 accordingly. Therefore, in the present disclosure, it is to be appreciated that the feedback voltage V_{fb} may be proportional to the regulated voltage V_{reg} .

[0039] In the examples in FIGS. **1** and **2**, the pass NFET **115** implements the pass transistor of the LDO regulator **100** regulator. The pass NFET **115** may be fabricated using a planar process, a FinFET process, and/or another fabrication process. Implementing the pass transistor of the LDO regulator **100** with the pass NFET **115** has several advantages over implementing the pass transistor with a p-type field effect transistor (PFET). One advantage is that the relatively low impedance of the pass NFET **115** at the output **130** of the LDO regulator **100** helps prevent a low-frequency pole from forming at the output **130**. This may eliminate the need for a large compensation capacitor at the output **130**. In addition, this may make the stability of the loop substantially independent of the load. Further, an NFET based LDO regulator may have better power supply rejection ratio (PSRR), low output ripple, and high stability.

[0040] However, a challenge with using the pass NFET **115** for the pass transistor of the LDO regulator **100** is that the gate voltage of the pass NFET **115** needs to be above the regulated voltage V_{reg} by at least the threshold voltage of the pass NFET **115** in order for the pass NFET **115** to turn on. In cases where the amplifier **120** is powered by the supply voltage V_{CC} , the maximum voltage that the amplifier **120** is able to output to the gate of the pass NFET **115** may be approximately V_{CC} . In these cases, the maximum regulated voltage V_{reg} at the output **130** of the LDO regulator **100** is limited to a voltage approximately equal to the supply voltage V_{CC} minus the threshold voltage of the pass NFET **115**. This prevents the regulated voltage V_{reg} from being set close to the supply voltage V_{CC} .

[0041] One approach to raise the regulated voltage V_{reg} closer to the supply voltage V_{CC} is to boost the voltage at the output **126** of the amplifier **120** and apply the boosted voltage to the gate of

the pass NFET **115**. In this regard, FIG. 3 shows an example where the LDO regulator **100** includes a voltage booster **330** coupled between the output **126** of the amplifier **120** and the gate of the pass NFET **115**. The voltage booster **330** has an input **332** coupled to the output **126** of the amplifier **120**, and an output **334** coupled to the gate of the pass NFET **115**. The input **332** may also be referred to as a boost input.

[0042] The voltage booster **330** is configured to receive the output voltage of the amplifier **120** at the input **332** of the voltage booster **330** (denoted “Vin”), boost (i.e., increase) the output voltage of the amplifier **120** to generate a boosted voltage, and output the boosted voltage at the output **334** of the voltage booster **330** (denoted “Vout”). For example, the voltage booster **330** may be configured to double the voltage at the output **126** of the amplifier **120**. In this example, the voltage booster **330** may be referred to as a voltage doubler. The boosted voltage at the gate of the pass NFET **115** may exceed the supply voltage VCC, allowing the LDO regulator **100** to set the regulated voltage Vreg closer to the supply voltage VCC and reduce the dropout voltage across the pass NFET **115**.

[0043] In the example in FIG. 3, the regulated voltage Vreg is fed directly to the second input **124** of the amplifier **120** via feedback path **150**. However, it is to be appreciated that the present disclosure is not limited to this example. For example, in some implementations, the feedback path **150** may include the voltage divider **215** shown in FIG. 2, in which case the feedback voltage Vfb is proportional to the regulated voltage Vreg based on the ratio of the resistances of the resistors R1 and R2, as discussed above.

[0044] The voltage booster **330** may be implemented with a charge pump or another type of circuit. In this regard, FIG. 4 shows an example in which the voltage booster **330** is implemented with a charge pump including a first switch **420**, a second switch **430**, a capacitor **410**, a control circuit **450**, and an output capacitor **460**. As discussed further below, the control circuit **450** controls the on/off states of the switches **420** and **430** via respective control inputs **425** and **435**. As used herein, a “control input” of a switch is an input that controls whether the switch is closed or open based on a signal (e.g., voltage) input to the control input. Each of the switches **420** and **430** may be implemented with a transistor, a transmission gate, or another type of switch. For the example of a switch implemented with a respective transistor, the respective control input corresponds to the gate of the respective transistor.

[0045] In this example, the first switch **420** is coupled between the input **332** of the voltage booster **330** and a first terminal **412** of the capacitor **410**, and the second switch **430** is coupled between the first terminal **412** of the capacitor **410** and the output **334** of the voltage booster **330**. The control circuit **450** is coupled to a second terminal **414** of the capacitor **410**. As discussed further below, the control circuit **450** controls the voltage applied to the second terminal **414** of the capacitor **410**. The output capacitor **460** may be coupled between the output **334** of the voltage booster **330** and ground. The output capacitor **460** may include multiple capacitors (e.g., coupled in parallel and/or another configuration).

[0046] In operation, the control circuit **450** receives a clock signal Clk at a clock input **455**, and times operations of the voltage booster **330** based on the clock signal Clk. The clock signal Clk may come from a phase locked loop (PLL) or another type of clock generator. During each period (i.e., cycle) of the clock signal Clk, the control circuit **450** charges the capacitor **410** during a charge phase (e.g., first portion of the clock period), and transfers charge between the capacitor **410** to the output capacitor **460** during a transfer phase (e.g., second portion of the clock period).

[0047] During the charge phase, the control circuit **450** turns on the first switch **420**, turns off the second switch **430**, and couples the second terminal **414** of the capacitor **410** to ground (i.e., 0V). This allows charge to flow from the output **126** of the amplifier **120** to the first terminal **412** of the capacitor **410** through the first switch **420**, charging the first terminal **412** of the capacitor **410** to the input voltage Vin.

[0048] During the transfer phase, the control circuit **450** turns off the first switch **420**, turns on the second switch **430**, and couples the voltage Vin at the input **332** to the second terminal **414** of the

capacitor **410**. The voltage V_{in} at the second terminal **414** of the capacitor **410** boosts the voltage at the first terminal **412** of the capacitor **410** to a voltage of $2V_{in}$ (i.e., double the input voltage V_{in}). The turning on of the second switch **430** allows the transfer of charge between the capacitor **410** and the output capacitor **460** through the second switch **430**. In the steady-state condition, the output voltage V_{out} is approximately equal to $2V_{in}$ in this example.

[0049] In the example discussed above, the control circuit **450** applies a voltage of V_{in} to the second terminal **414** of the capacitor **410** to boost at the voltage at the first terminal **412** of the capacitor **410** to $2V_{in}$. However, it is to be appreciated that the present disclosure is not limited to this example. In general, the control circuit **450** applies a boost voltage to the second terminal **414** of the capacitor **410** during the transfer phase to boost the voltage at the first terminal **412** of the capacitor **410** to V_{in} plus the boost voltage. In the above example, the boost voltage is V_{in} .

[0050] In the example shown in FIG. 4, a bleed resistor R_{bleed} is coupled between the output **130** of the LDO regulator **100** and ground. In this example, the bleed resistor R_{bleed} is configured to draw a small quiescent current that allows the LDO regulator **100** to maintain voltage regulation when the load current is zero or close to zero. It is to be appreciated that the bleed resistor R_{bleed} may be omitted in some implementations.

[0051] FIG. 5 shows an example in which the voltage booster **330** employs a double charge pumping architecture according to certain aspects. In this example, charge pumping is performed twice per period (i.e., cycle) of the clock signal Clk using two capacitors, as discussed further below.

[0052] In this example, the voltage booster **330** includes the first switch **420**, the second switch **430**, the capacitor **410**, and the control circuit **450** discussed above. In addition, the voltage booster **330** includes a third switch **520**, a fourth switch **530**, and a second capacitor **510**. In the discussion below, the capacitor **410** is referred to as the first capacitor.

[0053] The third switch **520** is coupled between the input **332** of the voltage booster **330** and a first terminal **512** of the second capacitor **510**, and the fourth switch **530** is coupled between first terminal **512** of the second capacitor **510** and the output **334** of the voltage booster **330**. The control circuit **450** is coupled to a second terminal **514** of the second capacitor **510**. In the example shown in FIG. 5, the control input **425** of the first switch **420** and the control input **525** of the third switch **520** are coupled to the control circuit **450**. The control input **435** of the second switch **430** may be coupled to the control circuit **450** (shown in the example in FIG. 5) or coupled to the first terminal **512** of the second capacitor **510**. The control input **535** of the fourth switch **530** may be coupled to the control circuit **450** (shown in the example in FIG. 5) or coupled to the first terminal **412** of the first capacitor **410**.

[0054] In operation, the control circuit **450** receives the clock signal Clk at the clock input **455**, and times operations of the voltage booster **330** based on the clock signal Clk . During each period (i.e., cycle) of the clock signal Clk , the control circuit **450** charges the first capacitor **410** and transfers charge between the second capacitor **510** and the output capacitor **460** during a first portion of the clock period, and charges the second capacitor **510** and transfers charge between the first capacitor **410** and the output capacitor **460** during a second portion of the clock period.

[0055] More particularly, during the first portion of the clock period, the control circuit **450** turns on the first switch **420**, turns off the second switch **430**, and couples the second terminal **414** of the first capacitor **410** to ground (i.e., $0V$). This allows the amplifier **120** to charge the first terminal **412** of the first capacitor **410** to the input voltage V_{in} through the first switch **420**. Also, during the first portion of the clock period, the control circuit **450** turns off the third switch **520**, turns on the fourth switch **530**, and couples the second terminal **514** of the second capacitor **510** to the boost voltage (e.g., V_{in}) to boost the voltage at the first terminal **512** of the second capacitor **510**. Alternately, the fourth switch **530** may be turned on by the voltage at the first terminal **412** of the first capacitor **410**, as discussed further below with reference to FIG. 6. The turning on of the fourth switch **530** allows the transfer of charge between the first terminal **512** of the second capacitor **510**

and the output capacitor **460**.

[0056] During the second portion of the clock period, the control circuit **450** turns off the first switch **420**, turns on the second switch **430**, and couples the second terminal **414** of the first capacitor **410** to the boost voltage (e.g., V_{in}) to boost the voltage at the first terminal **412** of the first capacitor **410**. Alternately, the second switch **430** may be turned on by the voltage at the first terminal **512** of the second capacitor **510**, as discussed further below with reference to FIG. 6. The turning on of the second switch **430** allows the transfer of charge between the first terminal **412** of the first capacitor **410** and the output capacitor **460**. Also, during the second portion of the clock period, the control circuit **450** turns on the third switch **520**, turns off the fourth switch **530**, and couples the second terminal **514** of the second capacitor **510** to ground (i.e., 0V). This allows the amplifier **120** to charge the first terminal **512** of the second capacitor **510** to the input voltage V_{in} through the third switch **520**.

[0057] Thus, in this example, the voltage booster **330** alternately charges the first capacitor **410** and the second capacitor **510** to the input voltage V_{in} . The voltage booster **330** also alternately transfers charge between the first capacitor **410** and the output capacitor **460** and between the second capacitor **510** and the output capacitor **460**. This allows the voltage booster **330** to perform two charge pump operations per clock period (i.e., cycle) using the capacitor **410** and the second capacitor **510**.

[0058] FIG. 6 shows an exemplary implementation of the first switch **420**, the second switch **430**, the third switch **520**, and the fourth switch **530**. In this example, the first switch **420** is implemented with a first n-type field effect transistor (NFET), in which control input **425** of the first switch **420** (located at the gate of the first NFET) receives the control signal $bst1$ from the control circuit **450**. The third switch **520** is implemented with a second NFET, in which control input **525** of the third switch **520** (located at the gate of the second NFET) receives the control signal $bst2$ from the control circuit **450**.

[0059] In this example, the second switch **430** is implemented with a first p-type field effect transistor (PFET), in which the control input **435** of the second switch **430** (located at the gate of the first PFET) is coupled to the first terminal **512** of the second capacitor **510**. The voltage at the first terminal **512** of the second capacitor **510** is labeled “ $\phi2_bst$ ” in FIG. 6. For ease of illustration, the connection between the control input **435** of the second switch **430** and the first terminal **512** of the second capacitor **510** is not shown in FIG. 6.

[0060] The fourth switch **530** is implemented with a second PFET, in which the control input **535** of the fourth switch **530** (located at the gate of the second PFET) is coupled to the first terminal **412** of the first capacitor **410**. The voltage at the first terminal **412** of the first capacitor **410** is labeled “ $\phi1_bst$ ” in FIG. 6. For ease of illustration, the connection between the control input **535** of the fourth switch **530** and the first terminal **412** of the first capacitor **410** is not shown in FIG. 6.

[0061] In FIG. 6, the control circuit **450** outputs the control signal $\phi1$ to the second terminal **414** of the first capacitor **410** and outputs the control signal $\phi2$ to the second terminal **514** of the second capacitor **510**, as discussed further below.

[0062] Exemplary operations of the voltage booster **330** will now be described with reference to FIG. 7 according to certain aspects. FIG. 7 is a timing diagram showing an example of the control signals $bst1$, $bst2$, $\phi1$, and $\phi2$ generated by the control circuit **450**. In this example, the control signals $bst1$, $bst2$, $\phi1$, and $\phi2$ are periodic signals having the same frequency (e.g., the frequency of the clock signal Clk). Each of the control signals $\phi1$ and $\phi2$ swings between the input voltage V_{in} of the voltage booster **330** and approximately zero volts. Each of the control signals $bst2$ and $bst1$ swings between V_{in} and $V_{in}+V_{CC}$.

[0063] FIG. 7 also shows an example of the voltage $\phi1_bst$ at the first terminal **412** of the first capacitor **410**, and the voltage $\phi2_bst$ at the first terminal **512** of the second capacitor **510**. In the example in FIG. 7, each of the voltages $\phi1_bst$ and $\phi2_bst$ swings between approximately V_{in} and $2V_{in}$. For ease of discussion, FIG. 7 shows the voltages $\phi1_bst$ and $\phi2_bst$ for the case

wherein the LDO regulator **100** is in a steady-state condition.

[0064] During time period **T1**, charge is transferred between the second capacitor **510** and the output capacitor **460**. More particularly, during the time period **T1**, the control circuit **450** sets the control signal ϕ_2 to V_{in} to apply a boost voltage of approximately V_{in} to the second terminal **514** of the second capacitor **510**. In the example in FIG. 7, this boosts the voltage ϕ_{2_bst} at the first terminal **512** of the second capacitor **510** to approximately $2V_{in}$. Also, during the time period **T1**, the control circuit **450** turns off (i.e., opens) the third switch **520** by setting control signal ϕ_2 to approximately V_{in} , and the fourth switch **530** is turned on (i.e., closed) by the voltage ϕ_{1_bst} at the first terminal **412** of the first capacitor **410**. This allows charge to flow between the second capacitor **510** and the output capacitor **460** through the fourth switch **530** while the voltage of the second capacitor **510** is boosted.

[0065] Also, during the time period **T1**, the control circuit **450** turns on the first switch **420** by setting control signal ϕ_1 to approximately $V_{in}+V_{CC}$, allowing charge to flow from the output **126** of the amplifier **120** to the first capacitor **410** to charge the first terminal **412** of the first capacitor **410** to V_{in} . During this time, the second switch **430** is turned off by the voltage ϕ_{2_bst} at the first terminal **512** of the second capacitor **510**.

[0066] Thus, during time period **T1**, charge is transferred between the second capacitor **510** and the output capacitor **460**, and the first terminal **412** of the first capacitor **410** is charged to V_{in} .

[0067] During time period **T2**, charge is transferred between the first capacitor **410** and the output capacitor **460**. More particularly, during the time period **T2**, the control circuit **450** sets the control signal ϕ_1 to V_{in} to apply a boost voltage of approximately V_{in} to the second terminal **414** of the first capacitor **410**. In the example in FIG. 7, this boosts the voltage ϕ_{1_bst} at the first terminal **412** of the first capacitor **410** to approximately $2V_{in}$. Also, during the time period **T2**, the control circuit **450** turns off (i.e., opens) the first switch **420** by setting control signal ϕ_1 to approximately V_{in} , and the second switch **430** is turned on (i.e., closed) by the voltage ϕ_{2_bst} at the first terminal **512** of the second capacitor **510**. This allows charge to flow between the first capacitor **410** and the output capacitor **460** through the second switch **430** while the voltage of the first voltage is boosted.

[0068] Also, during the time period **T2**, the control circuit **450** turns on the third switch **520** by setting control signal ϕ_2 to approximately $V_{in}+V_{CC}$, allowing charge to flow between the output **126** of the amplifier **120** to the second capacitor **510** to charge the first terminal **512** of the second capacitor **510** to V_{in} . During this time, the fourth switch **530** is turned off by the voltage ϕ_{1_bst} at the first terminal **412** of the first capacitor **410**.

[0069] Thus, during time period **T2**, charge is transferred between the first capacitor **410** and the output capacitor **460**, and the first terminal **512** of the second capacitor **510** is charged to V_{in} .

[0070] In the example shown in FIG. 6, the voltage booster **330** also includes a PFET **470** for reducing the startup time of the voltage booster **330**. The source of the PFET **470** is coupled to the input **332** of the voltage booster **330**, and the gate and drain of the PFET **470** are coupled to the output **334** of the voltage booster **330**. During startup when the input voltage V_{in} is greater than the output voltage V_{out} by at least the threshold voltage of the PFET **470**, the PFET **470** turns on and provides a charging path from the input **332** to the output **334** to quickly charge the output capacitor **460**. During the steady-state condition where the output voltage V_{out} is greater than the input voltage V_{in} (e.g., V_{out} is approximately $2V_{in}$), the PFET **470** is turned off.

[0071] FIG. 8 shows an exemplary implementation of the control circuit **450** according to certain aspects of the present disclosure. Note that the PFET **470** is not shown in FIG. 8 for ease of illustration.

[0072] As discussed above, the control circuit **450** generates control signals ϕ_1 and ϕ_2 for controlling the first switch **420** and the third switch **520**, respectively. In the example shown in FIG. 8, the control circuit **450** includes a first NFET **810**, a second NFET **815**, a third capacitor **820**, a fourth capacitor **830**, and a voltage boosting circuit **835**. The drains of the first and second

NFETs **810** and **815** are coupled to the input **332** of the voltage booster **330**. The first and second NFETs **810** and **815** are cross-coupled in which the gate of the first NFET **810** is coupled to the source of the second NFET **815**, and the gate of the second NFET **815** is coupled to the source of the first NFET **810**. A first terminal **822** of the third capacitor **820** is coupled to the source of the first NFET **810**, and a first terminal **832** of the fourth capacitor **830** is coupled to the source of the second NFET **815**.

[0073] In this example, the voltage boosting circuit **835** includes a signal generator **840**. The signal generator **840** has an input **842** configured to receive the clock signal Clk, a first output **844** coupled to a second terminal **824** of the third capacitor **820**, and a second output **846** coupled to a second terminal **834** of the fourth capacitor **830**. The signal generator **840** is configured to generate control signals phi1b and phi2b, output the signal phi1b to the second terminal **824** of the third capacitor **820** via the first output **844**, and output the signal phi2b to the second terminal **834** of the fourth capacitor **830** via the second output **846**. FIG. **9** shows an exemplary timing diagram of signals phi1b and phi2b, in which each of the signals phi1b and phi2b swings between VCC and approximately zero volts.

[0074] The control signal bst1 is taken from the first terminal **822** of the third capacitor **820**. Thus, in this example, the voltage of the control signal bst1 is equal to the voltage at the first terminal **822** of the third capacitor **820**. The control signal bst2 is taken from the first terminal **832** of the fourth capacitor **830**. Thus, in this example, the voltage of the control signal bst2 is equal to the voltage at the first terminal **832** of the fourth capacitor **830**.

[0075] During operation, the voltages of the signals phi1b and phi2b alternately swing to VCC. When the voltage of signal phi1b is VCC and the voltage of signal phi2b is low (e.g., approximately zero volts), the first NFET **810** is turned off and the second NFET **815** is turned on. The voltage at the first terminal **822** of the third capacitor **820** (and hence the voltage of control signal bst1) is boosted to a voltage approximately equal to the sum of Vin and VCC (i.e., Vin+VCC). As a result, the first switch **420** is turned on. The boosted voltage at the first terminal **822** of the third capacitor **820** (which is also coupled to the gate of the second NFET **815**) turns on the second NFET **815**. As a result, the fourth capacitor **830** is charged to Vin through the second NFET **815**.

[0076] When the voltage of signal phi1b is low (e.g., approximately zero volts) and the voltage of signal phi2b is VCC, the first NFET **810** is turned on and the second NFET **815** is turned off. The voltage at the first terminal **832** of the fourth capacitor **830** (and hence the voltage of control signal bst2) is boosted to a voltage approximately equal to Vin+VCC, which turns on the third switch **520**. The boosted voltage at the first terminal **832** of the fourth capacitor **830** (which is also coupled to the gate of the first NFET **810**) also turns on the first NFET **810**. As a result, the third capacitor **820** is charged to Vin through the first NFET **810**.

[0077] In the example in FIG. **8**, the voltage boosting circuit **835** also includes a first inverter **850** coupled between the first output **844** of the signal generator **840** and the second terminal **414** of the first capacitor **410**, and a second inverter **855** coupled between the second output **846** of the signal generator **840** and the second terminal **514** of the second capacitor **510**. In this example, the first inverter **850** is configured to generate signal phi1 by inverting signal phi1b from the signal generator **840**, and the second inverter **855** is configured to generate signal phi2 by inverting the signal phi2b from the signal generator **840**. In certain aspects, each of the inverters **850** and **855** is powered by the input voltage Vin of the voltage booster **330** so that the voltage of each of the signals phi1 and phi2 swings between Vin and approximately zero volts. For ease of illustration, the connections between the inverters **850** and **855** and the input **332** of the voltage booster **330** are not shown in FIG. **8**.

[0078] FIG. **9** is a timing diagram showing exemplary waveforms for the clock signal Clk and the controls signals phi1b, phi2b, phi1, and phi2 in the voltage booster of FIG. **8** according to certain aspects of the present disclosure. In the example shown in FIG. **9**, the high phases (e.g., VCC) of

the control signals ϕ_{1b} and ϕ_{2b} are non-overlapping.

[0079] During startup or a load transient, the output **126** of the amplifier **120** may saturate (i.e., reach VCC) due to the high input differential voltage (i.e., high difference between V_{ref} and V_{fb}) and the high gain of the amplifier **120**. In addition, voltages of transistors in the voltage booster **330** may get closer to limits of V_{dg} and V_{ds} . As a result, the output **126** of the amplifier **120** may not be increased further to speed up the settling of the LDO regulator **100**. This may cause the settling time of the LDO regulator **100** to be too long (e.g., exceed a settling time specification of the LDO regulator **100**). For the case of a load transient, the settling time may also be referred to as recovery time.

[0080] To address the above, aspects of the present disclosure provide a detection circuit configured to detect the startup or a load transient, and to switch the voltage booster **330** to a higher clock frequency when startup or the load transient is detected. The higher clock frequency increases the charge pumping speed of the voltage booster **330**, which reduces the settling time of the LDO regulator **100**. When the LDO regulator **100** settles to a steady state of operation, the detection circuit may switch the voltage booster **330** to a lower clock frequency to reduce the power consumption of the voltage booster **330**. The above features and other features of the present disclosure are discussed further below.

[0081] FIG. **10** shows an example in which the LDO regulator **100** includes a detection circuit **1020** and a multiplexer **1010** according to certain aspects.

[0082] In this example, the multiplexer **1010** has a first input **1012**, a second input **1014**, an output **1016**, and a select input **1018**. The first input **1012** is configured to receive a first clock signal $Clk1$, the second input **1014** is configured to receive a second clock signal $Clk2$, and the output **1016** is coupled to the clock input **455** of the voltage booster **330**. The multiplexer **1010** is configured to receive a clock select signal (labeled “Clk Sel”) from the detection circuit **1020**, select the first clock signal $Clk1$ or the second clock signal $Clk2$ based on the clock select signal, and output the selected clock signal to the clock input **455** of the voltage booster **330**.

[0083] In certain aspects, the first clock signal $Clk1$ has a clock frequency of f_a and the second clock signal $Clk2$ has a higher clock frequency of $Nf_{sub.d}$. Thus, in this example, the clock frequency of the second clock signal $Clk2$ is N times higher than the clock frequency of the first clock signal $Clk1$, where N is greater than one. The multiplexer **1010** allows the detection circuit **1020** to switch the voltage booster **330** to a higher clock frequency during startup or a load transient by causing the multiplexer **1010** to switch from the first clock signal $Clk1$ to the second clock signal $Clk2$ using the clock select signal. The clock signals $Clk1$ and $Clk2$ may be generated by a circuit (not shown) including one or more PLLs, one or more frequency dividers, or any combination thereof.

[0084] The detection circuit **1020** has an input **1022** coupled to the amplifier **120**, and an output **1024** coupled to the select input **1018** of the multiplexer **1010**. It is to be appreciated that the input **1022** may include one or more inputs. In certain aspects, the detection circuit **1020** is configured to detect startup or a load transient based on signals in the amplifier **120**, and cause the multiplexer **1010** to select the second clock signal $Clk2$ when the startup or the load transient is detected in order to switch the voltage booster **330** to the higher clock frequency $Nf_{sub.d}$. The higher clock frequency $Nf_{sub.d}$ increases the charge pumping speed of the voltage booster **330**, which reduces the settling time of the LDO regulator **100**. When the LDO regulator **100** settles to a steady state of operation, the detection circuit **1020** causes the multiplexer **1010** to select the first clock signal $Clk1$ to switch the voltage booster **330** to the lower clock frequency $f_{sub.d}$ to reduce the power consumption of the voltage booster **330** in steady state.

[0085] In certain aspects, the detection circuit **1020** is configured to detect the input differential voltage (i.e., difference between V_{ref} and V_{fb}) of the amplifier **120** based on signals in the amplifier **120** indicative of the input differential voltage. In these aspects, the detection circuit **1020** may be configured to cause the multiplexer **1010** to select the second clock signal $Clk2$ when the

input differential voltage is above a threshold voltage and cause the multiplexer **1010** to select the first clock signal Clk1 when the input differential voltage is below the threshold voltage.

[0086] For example, during startup, the feedback voltage Vfb may start at zero volts, resulting in a high input differential voltage (i.e., high difference between Vref and Vfb) that is above the threshold voltage. In this example, the detection circuit **1020** causes the multiplexer **1010** to select the second clock signal Clk2 to increase the clock frequency of the voltage booster **330**. The increased clock frequency reduces the settling time of the LDO regulator **100**, as discussed above. [0087] As the LDO regulator **100** settles, the feedback voltage Vfb increases, which decreases the input differential voltage (i.e., difference between Vref and Vfb). Eventually, the input differential voltage drops below the threshold, and the detection circuit **1020** causes the multiplexer **1010** to select the first clock signal Clk1 to reduce the clock frequency for steady state operation.

[0088] During a load transient, the load current may abruptly change by a large amount, resulting in a high input differential voltage (i.e., high difference between Vref and Vfb) that is above the threshold voltage. In this example, the detection circuit **1020** causes the multiplexer **1010** to switch from the first clock signal Clk1 to the second clock signal Clk2 to increase the clock frequency of the voltage booster **330**. The increased clock frequency reduces the recovery time of the LDO regulator **100**, as discussed above.

[0089] As the LDO regulator **100** recovers, the input differential voltage (i.e., difference between Vref and Vfb) decreases. Eventually, the input differential voltage drops below the threshold, and the detection circuit **1020** causes the multiplexer **1010** to switch back to the first clock signal Clk1. [0090] It is to be appreciated that the threshold voltage may vary across temperature and/or process corners. In this example, the detection circuit **1020** may be configured such that the threshold voltage stays above the input differential voltage during steady state operation across a range of temperatures and/or across process corners. In other words, it is to be appreciated the threshold voltage need not be fixed as long as the threshold voltage stays above the input differential voltage during steady state operation so that the detection circuit **1020** keeps the voltage booster **330** at the lower clock frequency fa during steady state operation.

[0091] FIG. **11** shows an example in which the detection circuit **1020** is coupled directly to the inputs **122** and **124** of the amplifier **120** to detect the input differential voltage. In this example, the input **1022** of the detection circuit **1020** includes a first input **1022-1** coupled to the first input **122** of the amplifier **120**, and a second input **1022-2** coupled to the second input **124** of the amplifier **120**. In this example, the detection circuit **1020** may detect the input differential voltage by sensing the voltage difference between the first input **1022-1** and the second input **1022-2**.

[0092] FIG. **12** shows an example implementation of the amplifier **120** and the detection circuit **1020** according to certain aspects of the present disclosure. In this example, the amplifier **120** includes an input stage **1210** (also referred to as a first stage) and a current mirror **1220**. The input stage **1210** includes a first input transistor **1222**, a second input transistor **1224**, and a bias current source **1216**. The first input transistor **1222** has a gate coupled to the first input **122** of the amplifier **120** to receive the reference voltage Vref, and a source coupled to the bias current source **1216**. The second input transistor **1224** has a gate coupled to the second input **124** of the amplifier **120**, and a source coupled to the bias current source **1216**. In the example shown in FIG. **12**, each of the input transistors **1222** and **1224** is implemented with a respective NFET. However, it is to be appreciated that the present disclosure is not limited to this example.

[0093] The bias current source **1216** is coupled between the sources of the input transistors **1222** and **1224** and ground (or some reference potential). The bias current source **1216** is configured to provide the input transistors **1222** and **1224** with a bias current. The current mirror **1220** is coupled to the drain of the first input transistor **1222**, the drain of the second input transistor **1224**, and the output **126** of the amplifier **120**.

[0094] In operation, the first input transistor **1222** is configured to convert the reference voltage Vref into a first current I.sub.1 at the drain of the first input transistor **1222**, and the second input

transistor **1224** is configured to convert the feedback voltage V_{fb} into a second current $I_{sub.2}$ at the drain of the second input transistor **1224**. In this regard, the input stage **1210** may also be referred to as a transconductance stage.

[0095] The current mirror **1220** is configured to generate a pull-up current $I_{sub.up_a}$ at the output **126** of the amplifier **120** based on the first current $I_{sub.1}$ by mirroring the first current $I_{sub.1}$ to the output **126**. The pull-up current $I_{sub.up_a}$ is proportional to the first current $I_{sub.1}$. The current mirror **1220** is also configured to generate a pull-down current $I_{sub.dn_a}$ at the output **126** of the amplifier **120** based on the second current $I_{sub.2}$ by mirroring the second current $I_{sub.2}$ to the output **126**. The pull-down current $I_{sub.dn_a}$ is proportional to the second current $I_{sub.2}$. As used herein, the term “proportional” converts the possibility of a proportionality factor (i.e., scaling factor) of one or greater than one.

[0096] When the pull-up current $I_{sub.up_a}$ is greater than the pull-down current $I_{sub.dn_a}$, current flows from the output **126** of the amplifier **120** to the capacitive load coupled to the output **126** of the amplifier **120**, which increases the input voltage V_{in} . The current flowing from the output **126** may be equal to the difference between the pull-up current $I_{sub.up_a}$ and the pull-down current $I_{sub.dn_a}$. In this example, the pull-up current $I_{sub.up_a}$ may be greater than the pull-down current $I_{sub.dn_a}$ when V_{ref} is greater than V_{fb} . The capacitive load may include a capacitor (not shown) coupled to the output **126**, parasitic capacitances, capacitances in the voltage booster **330**, or any combination thereof.

[0097] When the pull-down current $I_{sub.dn_a}$ is greater than the pull-up current $I_{sub.up_a}$, the output **126** of the amplifier **120** draws current from the capacitive load coupled to the output **126** of the amplifier **120**, which decreases the input voltage V_{in} . The current drawn by the output **126** may be equal to the difference between the pull-down current $I_{sub.dn_a}$ and the pull-up current $I_{sub.up_a}$. In this example, the pull-down current $I_{sub.dn_a}$ may be greater than the pull-up current $I_{sub.up_a}$ when V_{fb} is greater than V_{ref} .

[0098] In the example in FIG. **12**, the current mirror **1220** includes a first transistor **1226**, a second transistor **1228**, a third transistor **1230**, a fourth transistor **1232**, a fifth transistor **1234**, and a sixth transistor **1236**. The source of the first transistor **1226** is coupled to the supply rail **112**, the drain of the first transistor **1226** is coupled to the drain of the first input transistor **1222**, and the gate and the drain of the first transistor **1226** are coupled together. The source of the second transistor **1228** is coupled to the supply rail **112**, the gate of the second transistor **1228** is coupled to the gate of the first transistor **1226**, and the drain of the second transistor **1228** is coupled to the output **126** of the amplifier **120**. In this example, the first transistor **1226** and the second transistor **1228** mirror the first current $I_{sub.1}$ to the drain of the second transistor **1228** to provide the pull-up current $I_{sub.up_a}$. The pull-up current $I_{sub.up_a}$ is proportional to the first current $I_{sub.1}$ depending on the ratio of the channel width of the first transistor **1226** to the channel width of the second transistor **1228**.

[0099] The source of the third transistor **1230** is coupled to the supply rail **112**, the drain of the third transistor **1230** is coupled to the drain of the second input transistor **1224**, and the gate and the drain of the third transistor **1230** are coupled together. The source of the fourth transistor **1232** is coupled to the supply rail **112**, and the gate of the fourth transistor **1232** is coupled to the gate of the third transistor **1230**. The drain of the fifth transistor **1234** is coupled to drain of the fourth transistor **1232**, the gate and the drain of the fifth transistor **1234** are coupled together, and the source of the fifth transistor **1234** is coupled to ground. The gate of the sixth transistor **1236** is coupled to the gate of the fifth transistor **1234**, the source of the sixth transistor **1236** is coupled to ground, and the drain to the sixth transistor **1236** is coupled to the output **126** of the amplifier **120**. In this example, the transistors **1230**, **1232**, **1234**, and **1236** mirror the second current $I_{sub.2}$ to the drain of the sixth transistor **1236** to provide the pull-down current $I_{sub.dn_a}$. The pull-down current $I_{sub.dn_a}$ is proportional to the second current $I_{sub.2}$ depending on relative channel widths of the transistors **1230**, **1232**, **1234**, and **1236**.

[0100] In the example shown in FIG. 12, each of the transistors 1226, 1228, 1230, and 1232 is implemented with a respective PFET, and each of the transistors 1234 and 1236 is implemented with a respective NFET. However, it is to be appreciated that the present disclosure is not limited to this example.

[0101] In the example in FIG. 12, the detection circuit 1020 includes a current mirror 1240 coupled to the drains of the input transistors 1222 and 1224, which are coupled to the gates of the transistors 1226 and 1230, respectively. In this example, the current mirror 1240 is configured to mirror the first current $I_{sub.1}$ to provide a pull-up current $I_{sub.up_d}$ at the output 1024 of the detection circuit 1020. The current mirror 1240 is also configured to mirror the second current $I_{sub.2}$ to provide a pull-down current $I_{sub.dn_d}$ at the output 1024 of the detection circuit 1020.

[0102] In this example, the pull-up current $I_{sub.up_d}$ is proportional to the first current $I_{sub.1}$ by a first scaling factor, and the pull-down current $I_{sub.dn_d}$ is proportional to the second current $I_{sub.2}$ by a second scaling factor. The second scaling factor is M times (e.g., ten times) the first scaling factor where M is greater than one. The reason for the larger scaling factor for the pull-down current $I_{sub.dn_d}$ is discussed below. The first scaling factor may be one or greater than one.

[0103] In this example, the current mirror 1240 is configured to pull the output 1024 to the supply voltage VCC (i.e., logic one) when the pull-up current $I_{sub.up_d}$ is greater than the pull-down current $I_{sub.dn_d}$, and pull the output 1024 to ground (i.e., logic zero) when the pull-down current $I_{sub.dn_d}$ is greater than the pull-up current $I_{sub.up_d}$. In the example shown in FIG. 12, the multiplexer 1010 is configured to select the first clock signal $Clk1$ when the output 1024 of the detection circuit 1020 is logic zero, and select the second clock signal $Clk2$ when the output 1024 of the detection circuit 1020 is logic one.

[0104] Exemplary operations of the detection circuit 1020 will now be described according to certain aspects.

[0105] When the input differential voltage (i.e., difference between V_{ref} and V_{fb}) is high, the second input transistor 1224 may be turned off or mostly turned off while the first input transistor 1222 may be strongly turned on. For example, during startup, the feedback voltage V_{fb} may initially be approximately zero volts, turning off the second input transistor 1224. As a result, the second current $I_{sub.2}$ (and hence the pull-down current $I_{sub.dn_d}$) may be zero or close to zero amps. This allows the pull-up current $I_{sub.up_d}$ to pull the output 1024 of the detection circuit 1020 to logic one, which causes the multiplexer 1010 to select the second clock signal $Clk2$ (i.e., the higher clock frequency). The higher clock frequency speeds up the charge pumping of the voltage booster 330, which reduces the settling time.

[0106] As the LDO regulator 100 settles to the steady state, the input differential voltage (i.e., difference between V_{ref} and V_{fb}) becomes smaller. When the feedback voltage V_{fb} gets close to the reference V_{ref} , the pull-down current $I_{sub.up_d}$ becomes greater than the pull-up current $I_{sub.dn_a}$ due to the larger scaling factor for the pull-down current $I_{sub.dn_d}$. As a result, the pull-down current $I_{sub.up_d}$ pulls the output 1024 of the detection circuit 1020 to logic zero (e.g., ground potential), which causes the multiplexer 1010 to select the first clock signal Clk (i.e., lower frequency clock) for steady state operation. In this example, the larger scaling factor for the pull-down current $I_{sub.dn_d}$ helps ensure that the pull-down current $I_{sub.dn_d}$ is greater than the pull-up current $I_{sub.up_d}$ during steady state operation even when the first current $I_{sub.1}$ is slightly greater than the second current $I_{sub.2}$ during steady state operation.

[0107] In this example, the threshold of the detection circuit 1020 discussed above depends on the scaling factor difference (i.e., M) between the pull-down current $I_{sub.up_d}$ and the pull-up current $I_{sub.up_d}$. For example, the threshold may be increased by increasing the scaling factor difference (i.e., M). This is because, when the scaling factor difference is increased, the first current $I_{sub.1}$ needs to be higher than the second current $I_{sub.2}$ by a greater amount in order for the pull-up current $I_{sub.dn_d}$ to be higher than the pull-down current $I_{sub.dn_d}$ to pull the output 1024 of the detection circuit 1020 to logic one.

[0108] In the example in FIG. 12, the current mirror **1240** includes a first transistor **1250**, a second transistor **1252**, a third transistor **1254**, and a fourth transistor **1256**. The source of the first transistor **1250** is coupled to the supply rail **112**, the gate of the first transistor **1250** is coupled to the first input **1022-1**, and the drain of the first transistor **1250** is coupled to the output **1024** of the detection circuit **1020**. In this example, the first input **1022-1** of the detection circuit **1020** is coupled to the drain of the first input transistor **1222** and the gate of the first transistor **1226** in the amplifier **120**. In this example, the first current $I_{sub.1}$ is mirrored at the drain of the first transistor **1250** to provide the pull-up current $I_{sub.dn_d}$.

[0109] The source of the second transistor **1252** is coupled to the supply rail **112**, and the gate of the second transistor **1252** is coupled to the second input **1022-2**. The second input **1022-2** is coupled to the drain of the second input transistor **1224** and the gate of the third transistor **1230** in the amplifier **120**. The drain of the third transistor **1254** is coupled to the drain of the second transistor **1252**, the drain and the gate of the third transistor **1254** are coupled together, and the source of the third transistor **1254** is coupled to ground. The gate of the fourth transistor **1256** is coupled to the gate of the third transistor **1254**, the drain of the fourth transistor **1256** is coupled to the output **1024** of the detection circuit **1020**, and the source of the fourth transistor **1256** is coupled to ground. In this example, the second current $I_{sub.2}$ is mirrored at the drain of the fourth transistor **1256** to provide the pull-down current $I_{sub.dn_d}$. In the example shown in FIG. 12, the channel widths of each of the second transistor **1252**, the third transistor **1254**, and the fourth transistor **1256** is M times the channel width of the first transistor **1250** such that the second scaling factor is M times the first scaling factor.

[0110] FIG. 13 illustrates a method **1300** of operating a voltage regulator (e.g., the LDO regulator **100**) according to certain aspects. The voltage regulator includes a pass transistor (e.g., the pass NFET **115**), an amplifier (e.g., the amplifier **120**), and a voltage booster (e.g., the voltage booster **330**) coupled between an output of the amplifier and a gate of the pass transistor, wherein a first input (e.g., first input **122**) of the amplifier receives a reference voltage and a second input (e.g., second input **124**) of the amplifier receives a feedback voltage via a feedback path (e.g., feedback path **150**) coupled to a source of the pass transistor.

[0111] At block **1310**, a difference between the reference voltage and the feedback voltage greater than a threshold voltage is detected. For example, the difference between the reference voltage and the feedback voltage may be detected by the detection circuit **1020**. In this example, the voltage difference between the reference voltage and the feedback voltage being greater than the threshold voltage may indicate startup of the voltage regulator or a load transient.

[0112] At block **1320**, a clock input of the voltage booster is switched from a first clock signal to a second clock signal in response to detecting the difference between the reference voltage and the feedback voltage is greater than the threshold voltage, wherein the second clock signal has a higher clock frequency than the first clock signal. For example, the clock input (e.g., clock input **455**) of the voltage booster may be switched from the first clock signal (e.g., first clock signal Clk1) to the second clock signal (e.g., second clock signal Clk2) by the multiplexer **1010**.

[0113] In certain aspects, the method **1300** may further include detecting the difference between the reference voltage and the feedback voltage is less than the threshold voltage, and switching the clock input of the voltage booster from the second clock signal to the first clock signal in response to detecting the difference between the reference voltage and the feedback voltage is less than the threshold voltage. In this example, the voltage difference between the reference voltage and the feedback voltage being less than the threshold voltage may indicate that the voltage regulator has settled to a steady state.

[0114] In the above examples, the detection circuit **1020** reduces settling time during startup or a load transient by switching the LDO regulator **100** to the higher clock frequency (e.g., the second clock signal) when the startup or the load transient is detected. In other implementations, the detection circuit **1020** reduces settling time during startup or a load transient by switching the boost

capacitor (e.g., capacitor **410**) to a higher capacitance when the startup or the load transient is detected. The higher capacitance reduces the settling time by increasing the total amount of charge transfer in a unit time (e.g., in a cycle of the clock signal Clk). When the LDO regulator **100** settles to a steady state of operation, the detection circuit **1020** may switch the boost capacitor to a lower capacitance. The above features and other features of the present disclosure are discussed further below.

[0115] FIG. **14** shows an exemplary implementation of the voltage booster **330** according to certain aspects. In this example, the voltage booster **330** includes the control circuit **450**, the first switch **420**, the second switch **430**, the capacitor **410**, and the output capacitor **460** discussed above. The voltage booster **330** also includes a capacitance controller **1420** configured to switch the capacitor **410** between a first capacitance and a second capacitance higher than the first capacitance. As discussed further below, the detection circuit **1020** causes the capacitance controller **1420** to switch the capacitor **410** from the first capacitance to the second capacitance when the detection circuit **1020** detects startup or a load transient. The capacitor **410** may also be referred to as a boost capacitor since the capacitor **410** is used for voltage boosting.

[0116] As discussed above, during the charge phase, the control circuit **450** charges the capacitor **410** with current from the amplifier **120** by closing the first switch **420**. The capacitor **410** may be charged to the voltage V_{in} at the input **332** of the voltage booster **330** (which is coupled to the output **126** of the amplifier **120**). During the transfer phase, the control circuit **450** boosts the voltage of the capacitor **410** (e.g., to a voltage of $2V_{in}$ for a voltage doubler implementation) and transfers charge from the capacitor **410** to the output capacitor **460** by closing the second switch **430**.

[0117] In the example in FIG. **14**, the input **1022** of the detection circuit **1020** is coupled to the amplifier **120** and the output **1024** of the detection circuit **1020** is coupled to the capacitance controller **1420**. The detection circuit **1020** may be implemented with any of the exemplary implementations discussed above with references to FIGS. **10**, **11** and **12**. In operation, the detection circuit **1020** is configured to detect startup or a load transient, and cause the capacitance controller **1420** to switch the capacitor **410** to the second capacitance (i.e., higher capacitance) when the startup or the load transient is detected.

[0118] As discussed above, the detection circuit **1020** may detect the startup or the load transient by detecting the input differential voltage (i.e., difference between V_{ref} and V_{fb}) of the amplifier **120** based on signals in the amplifier **120** indicative of the input differential voltage. The detection circuit **1020** may detect the startup or the load transient when the input differential voltage is greater than a threshold voltage. In this example, the detection circuit **1020** may cause the capacitance controller **1420** to switch the capacitor **410** from the first capacitance to the second capacitance when the input differential voltage is greater than the threshold voltage. The detection circuit **1020** may cause the capacitance controller **1420** to switch the capacitor **410** from the second capacitance to the first capacitor when the input differential voltage is less than the threshold voltage.

[0119] FIG. **15** shows an example in which the detection circuit **1020** is coupled directly to the inputs **122** and **124** of the amplifier **120** to detect the input differential voltage. In this example, the input **1022** of the detection circuit **1020** includes the first input **1022-1** coupled to the first input **122** of the amplifier **120**, and the second input **1022-2** coupled to the second input **124** of the amplifier **120**. In this example, the detection circuit **1020** may detect the input differential voltage by sensing the voltage difference between the first input **1022-1** and the second input **1022-2**.

[0120] In the examples shown in FIGS. **14** and **15**, the capacitance controller **1420** has an input **1425** coupled to the output **1024** of the detection circuit **1020**. The capacitance controller **1420** may be configured to switch the capacitor **410** to the first capacitance when the capacitance controller **1420** receives a first logic value (e.g., zero) at the input **1425** and switch the capacitor **410** to the second capacitance when the capacitance controller **1420** receives a second logic value (e.g., one)

at the input **1425**. In this example, the detection circuit **1020** causes the capacitance controller **1420** to switch the capacitor **410** to the second capacitance by outputting the second logic value (e.g., one) to the input **1425** of the capacitance controller **1420**, and causes the capacitance controller **1420** to switch the capacitor **410** to the first capacitance by outputting the first logic value (e.g., zero) to the input **1425** of the capacitance controller **1420**.

[0121] In the examples shown in FIGS. **14** and **15**, the capacitor **410** includes a first capacitor **1412** and a second capacitor **1414**, and the capacitance controller **1420** includes a switch **1430**. The top terminal of the first capacitor **1412** is coupled to the switches **420** and **430** and the bottom terminal of the first capacitor **1412** is coupled to the control circuit **450**. The second capacitor **1414** and the switch **1430** are coupled in series between the switches **420** and **430** and the control circuit **450**. In this example, the first capacitor **1412** has a capacitance of C and the second capacitor **1414** has a capacitance of nC where n may be an integer, a fraction, or a mixed number.

[0122] In this example, the switch **1430** is opened to switch the capacitor **410** to the first capacitance. In this case, the bottom terminal of the second capacitor **1414** is floating and the first capacitance is approximately equal to the capacitance C of the first capacitor **1412**. The switch **1430** is closed to switch the capacitor **410** to the second capacitance. In this case, the second capacitance is approximately equal to the sum of the capacitances of the first capacitor **1412** and the second capacitor **1414** (i.e., $C+nC$). In this example, the switch **1430** may be configured to open when the capacitance controller **1420** receives the first logic value (e.g., zero) at the input **1425** and close when the capacitance controller **1420** receives the second logic value (e.g., one) at the input **1425**. The switch **1430** may be implemented with a transistor, a transmission gate, or another type of switch. For the example where the switch **1430** is implemented with a transistor, the gate of the transistor is coupled to the input **1425** of the capacitance controller **1420**.

[0123] FIG. **16** shows an example in which the LDO regulator **100** also includes a pulse stretcher **1610** coupled between the output **1024** of the detection circuit **1020** and the input **1425** of the capacitance controller **1420**. The pulse stretcher **1610** has an input **1612** coupled to the output **1024** of the detection circuit **1020** and an output **1614** coupled to the input **1425** of the capacitance controller **1420**.

[0124] When the detection circuit **1020** outputs the second logic value (e.g., one) to switch the capacitor **410** to the second capacitance, the pulse stretcher **1610** passes the second logic value to the input **1425** of the capacitance controller **1420**. When the output **1024** of the detection circuit **1020** transitions from the second logic value (e.g., one) to the first logic value (e.g., zero), the output **1614** of the pulse stretcher **1610** transitions from the second logic value to the first logic value after a short delay. The delay causes the pulse stretcher **1610** to extend the amount of time that the second logic value (e.g., one) is output to the capacitance controller **1420** to sustain the fast response action (i.e., higher capacitance) a little longer without impacting the phase margin.

[0125] FIG. **17** shows an exemplary implementation of the pulse stretcher **1610** according to certain aspects. In this example, the pulse stretcher **1610** includes an NFET **1710**, a PFET **1720**, a resistor **1730**, a capacitor **1740**, and an inverter **1750**. The source of the PFET **1720** is coupled to the supply rail and the gate of the PFET **1720** is coupled to the input **1612**. The source of the NFET **1710** is coupled to ground and the gate of the NFET **1710** is coupled to the input **1612**. The resistor **1730** is coupled between the drain of the PFET **1720** and the drain of the NFET **1710**. The inverter **1750** is coupled between the drain of the NFET **1710** and the output **1614**, and the capacitor **1740** is coupled between the drain of the NFET **1710** and ground.

[0126] In this example, the first logic value is zero and the second logic value is one. When the output **1024** of the detection circuit **1020** transitions from one to zero to switch the capacitor **410** to the first capacitance, the capacitor **1740** and the resistor **1730** delay the transition from one to zero at the output **1614** of the pulse stretcher **1610**. This is because it takes time for the PFET **1720** to charge up the capacitor **1740** through the resistor **1730**. Since the capacitor **1740** is coupled to the input of the inverter **1750**, the time to charge up the capacitor **1740** delays the transition of the

output of the inverter **1750** from one to zero, and hence delays the transition at the output **1614** of the pulse stretcher **1610** from one to zero.

[0127] Aspects of the present disclosure may be extended to the exemplary double charge pumping architecture shown in FIGS. **5**, **6** and **8**. In the regard, FIG. **18** shows an example in which the second capacitor **510** includes a third capacitor **1812** and a fourth capacitor **1814**. The third capacitor **1812** may have approximately the same capacitance (e.g., C) as the first capacitor **1412**, and the fourth capacitor **1814** may have approximately the same capacitance (e.g., nC) as the second capacitor **1414**. However, it is to be appreciated that the present disclosure is not limited to this example.

[0128] In this example, the LDO regulator **100** includes a capacitance controller **1820** configured to switch the second capacitor **510** between a third capacitance and a fourth capacitance greater than the third capacitance. The third capacitance may be approximately equal to the first capacitance (e.g., C), and the fourth capacitance may be approximately equal to the second capacitance (e.g., C+nC). In the example shown in FIG. **18**, the capacitance controller **1820** includes a switch **1830** coupled in series with the fourth capacitor **1814**.

[0129] In this example, the top terminal of the third capacitor **1812** is coupled to the switches **520** and **530** and the bottom terminal of the third capacitor **1812** is coupled to the control circuit **450**. The fourth capacitor **1814** and the switch **1830** are coupled in series between the switches **520** and **530** and the control circuit **450**. In this example, the switch **1830** is opened to switch the second capacitor **510** to the third capacitance and closed to switch the second capacitor **510** to the fourth capacitance.

[0130] In this example, the detection circuit **1020** may cause the capacitance controllers **1420** and **1820** to switch the first capacitor **410** and the second capacitor **510** to the second capacitance and the fourth capacitance, respectively, when the detection circuit **1020** detects startup or a load transient. As discussed above, the detection circuit **1020** may detect the startup or the load transient by detecting the input differential voltage (i.e., difference between Vref and Vfb) of the amplifier **120** and detecting the startup or the load transient when the input differential voltage is greater than a threshold voltage. In the example in FIG. **18**, the detection circuit **1020** may switch the first capacitor **410** to the second capacitance (e.g., C+nC) by closing the switch **1430**, and switch the second capacitor **510** to the fourth capacitance (e.g., C+nC) by closing the switch **1830**.

[0131] When the input different voltage is less than the threshold voltage, the detection circuit **1020** may cause the capacitance controllers **1420** and **1820** to switch the first capacitor **410** and the second capacitor **510** to the first capacitance (e.g., C) and the third capacitance (e.g., C), respectively. For example, the detection circuit **1020** may open the switch **1430** to switch the first capacitor **410** to the first capacitance, and open the switch **1830** to switch the second capacitor **510** to the third capacitance.

[0132] Although not shown in FIG. **18**, it is to be appreciated that the LDO regulator **100** may include the pulse stretcher **1610** shown in FIG. **16**. In this example, input **1612** of the pulse stretcher **1610** is coupled to the output **1024** of the detection circuit **1020**, and the output **1614** of the pulse stretcher **1610** is coupled to the input **1425** of the capacitance controller **1420** for the first capacitor **410** and the input **1825** of the capacitance controller **1820** for the second capacitor **510**.

[0133] FIG. **19** shows an exemplary implementation in which the capacitance controller **1420** includes a driver **1910**. The output of the driver **1910** is coupled to the second capacitor **1414** and an enable/disable input of the driver **1910** is coupled to the input **1425** of the capacitance controller **1420**. In the example shown in FIG. **19**, the input of the driver **1910** is coupled to the first output **844** of the signal generator **840** and receives the control signal phi1b from the signal generator **840**. However, it is to be appreciated that the present disclosure is not limited to this example. In the example shown in FIG. **19**, the driver **1910** is implemented with a tri-state inverter. However, it is to be appreciated that the driver is not limited to this example. In this example, the driver **1910** may be disabled when the first logic value (e.g., zero) is input to the input **1425**, and enabled when the

second logic value (e.g., one) is input to the input **1425**. However, it is to be appreciated that the driver **1910** is not limited to this example.

[0134] In this example, the detection circuit **1020** may switch the capacitor **410** to the first capacitance by disabling the driver **1910** (e.g., inputting the first logic value (e.g., zero) to the input **1425**). In this case, the output impedance of the driver **1910** may be very high when the driver **1910** is disabled. The detection circuit **1020** may switch the capacitor **410** to the second capacitance by enabling the driver **1910** (e.g., inputting the second logic value (e.g., logic one) to the input **1425**). In this case, the first inverter **850** drives the first capacitor **1412** and the driver **1910** drives the second capacitor **1414**. For the example where the driver **1910** is implemented with a tri-state inverter, both the first inverter **850** and the driver **1910** invert the control signal $\phi 1b$ to drive the first capacitor **1412** and the second capacitor **1414**, respectively.

[0135] FIG. **19** shows an exemplary implementation in which the capacitance controller **1820** for the second capacitor **510** includes a driver **1920**. The output of the driver **1920** is coupled to the fourth capacitor **1814** and an enable/disable input of the driver **1920** is coupled to the input **1825** of the capacitance controller **1820**. In the example shown in FIG. **19**, the input of the driver **1920** is coupled to the second output **846** of the signal generator **840** and receives the control signal $\phi 2b$ from the signal generator **840**. However, it is to be appreciated that the present disclosure is not limited to this example. In the example shown in FIG. **19**, the driver **1920** is implemented with a tri-state inverter. However, it is to be appreciated that the driver is not limited to this example. In this example, the driver **1920** may be disabled when the first logic value (e.g., zero) is input to the input **1825**, and enabled when the second logic value (e.g., one) is input to the input **1825**.

[0136] In this example, the detection circuit **1020** may switch the second capacitor **510** to the third capacitance (e.g., C) by disabling the driver **1920**. In this case, the output impedance of the driver **1920** may be very high when the driver **1920** is disabled. The detection circuit **1020** may switch the second capacitor **510** to the fourth capacitance (e.g., $C+nC$) by enabling the driver **1920**. In this case, the second inverter **855** drives the third capacitor **1812** and the driver **1920** drives the fourth capacitor **1814**. For the example where the driver **1920** is implemented with a tri-state inverter, both the second inverter **855** and the driver **1920** invert the control signal $\phi 2b$ to drive the third capacitor **1812** and the fourth capacitor **1814**, respectively.

[0137] Although not shown in FIG. **19**, it is to be appreciated that the LDO regulator **100** may include the pulse stretcher **1610** shown in FIG. **16**. In this example, input **1612** of the pulse stretcher **1610** is coupled to the output **1024** of the detection circuit **1020**, and the output **1614** of the pulse stretcher **1610** is coupled to the input **1425** of the capacitance controller **1420** for the first capacitor **410** and the input **1825** of the capacitance controller **1820** for the second capacitor **510**.

[0138] FIG. **20** illustrates a method **2000** of operating a voltage regulator (e.g., the LDO regulator **100**) according to certain aspects. The voltage regulator includes a pass transistor (e.g., the pass NFET **115**), an amplifier (e.g., the amplifier **120**), and a voltage booster (e.g., the voltage booster **330**) coupled between an output of the amplifier and a gate of the pass transistor, wherein a first input (e.g., first input **122**) of the amplifier receives a reference voltage and a second input (e.g., second input **124**) of the amplifier receives a feedback voltage via a feedback path (e.g., feedback path **150**) coupled to a source of the pass transistor.

[0139] At block **2010**, a difference between the reference voltage and the feedback voltage greater than a threshold voltage is detected. For example, the difference between the reference voltage and the feedback voltage may be detected by the detection circuit **1020**. In this example, the voltage difference between the reference voltage and the feedback voltage being greater than the threshold voltage may indicate startup of the voltage regulator or a load transient.

[0140] At block **2020**, a boost capacitor of the voltage booster is switched from a first capacitance to a second capacitance in response to detecting the difference between the reference voltage and the feedback voltage is greater than the threshold voltage, wherein the second capacitance is higher than the first capacitance. For example, the boost capacitor may correspond to capacitor **410**, the

first capacitance may correspond to the capacitance C, and the second capacitance may correspond to the capacitance C+nC (e.g., the combined capacitance of the first capacitor **1412** and the second capacitor **1414**).

[0141] In certain aspects, the method **2000** may further include detecting the difference between the reference voltage and the feedback voltage is less than the threshold voltage, and switching the boost capacitor of the voltage booster from the second capacitance to the first capacitance in response to detecting the difference between the reference voltage and the feedback voltage is less than the threshold voltage.

[0142] Implementation examples are described in the following numbered clauses:

[0143] 1. A voltage regulator, comprising: [0144] a pass transistor coupled between an input of the voltage regulator and an output of the voltage regulator; [0145] an amplifier having a first input, a second input, and an output, wherein the first input of the amplifier is configured to receive a reference voltage, and the second input of the amplifier is coupled to the output of the voltage regulator via a feedback path; [0146] a voltage booster coupled between the output of the amplifier and a gate of the pass transistor; [0147] a multiplexer having a first input, a second input, an output, and a select input, wherein the first input of the multiplexer is configured to receive a first clock signal, the second input of the multiplexer is configured to receive a second clock signal having a higher frequency than the first clock signal, and the output of the multiplexer is coupled to a clock input of the voltage booster; and [0148] a detection circuit having an input and an output, wherein the input of the detection circuit is coupled to the amplifier, and the output of the detection circuit is coupled to the select input of the multiplexer.

[0149] 2. The voltage regulator of clause 1, wherein the voltage booster comprises a charge pump.

[0150] 3. The voltage regulator of clause 1 or 2, wherein the detection circuit is configured to:

[0151] detect a difference between the reference voltage at the first input of the amplifier and a feedback voltage at the second input of the amplifier; [0152] cause the multiplexer to select the second clock signal if the difference between the reference voltage and the feedback voltage is greater than a threshold voltage; and [0153] cause the multiplexer to select the first clock signal if the difference between the reference voltage and the feedback voltage is less than the threshold voltage.

[0154] 4. The voltage regulator of clause 3, wherein the input of the detection circuit includes a first input coupled to the first input of the amplifier and a second input coupled to the second input of the amplifier.

[0155] 5. The voltage regulator of clause 1 or 2, wherein: [0156] the amplifier comprises an input stage configured to: [0157] convert the reference voltage at the first input of the amplifier into a first current; and [0158] convert a feedback voltage at the second input of the amplifier into a second current; and [0159] the detection circuit comprises a current mirror configured to: [0160] generate a pull-up current at the output of the detection circuit that is proportional to the first current by a first scaling factor; and [0161] generate a pull-down current at the output of the detection circuit that is proportional to the second current by a second scaling factor greater than the first scaling factor.

[0162] 6. The voltage regulator of clause 5, wherein the multiplexer is configured to: [0163] select the second clock signal if the output of the detection circuit is logic one; and [0164] select the first clock signal if the output of the detection circuit is logic zero.

[0165] 7. The voltage regulator of clause 1 or 2, wherein amplifier comprises: [0166] a first input transistor, wherein a gate of the first input transistor is coupled to the first input of the amplifier, and the first input transistor is configured to convert the reference voltage at the first input of the amplifier into a first current; and [0167] a second input transistor, wherein a gate of the second input transistor is coupled to the second input of the amplifier, and the second input transistor is configured to convert a feedback voltage at the second input of the amplifier into a second current.

[0168] 8. The voltage regulator of clause 7, wherein the detection circuit comprises: [0169] a first

transistor, wherein a gate of the first transistor is coupled to a drain of the first input transistor, and a drain of the first transistor is coupled to the output of the detection circuit; [0170] a second transistor, wherein a gate of the second transistor is coupled to a drain of the second input transistor; [0171] a third transistor, wherein a drain of the third transistor is coupled to a drain of the second transistor, and a gate of the third transistor is coupled to the drain of the third transistor; and [0172] a fourth transistor, wherein a gate of the fourth transistor is coupled to the gate of the third transistor, a drain of the fourth transistor is coupled to the output of the detection circuit, a channel width of the fourth transistor is M times greater than a channel width of the first transistor, and M is greater than one.

[0173] 9. The voltage regulator of any one of clauses 1 to 8, wherein the voltage booster is configured to double a voltage at the output of the amplifier to obtain a boosted voltage, and output the boosted voltage to the gate of the pass transistor.

[0174] 10. The voltage regulator of any one of claims 1 to 9, wherein the voltage booster comprises: [0175] a capacitor having a first terminal and a second terminal; [0176] a first switch coupled between the output of the amplifier and the first terminal of the capacitor; [0177] a second switch coupled between the first terminal of the capacitor and the gate of the pass transistor; and [0178] a control circuit coupled to the clock input of the voltage booster, a control input of the first switch, a control input of the second switch, and the second terminal of the capacitor.

[0179] 11. The voltage regulator of clause 10, wherein the voltage booster further comprises an output capacitor coupled between the gate of the pass transistor and a ground.

[0180] 12. The voltage regulator of clause 10 or 11, wherein, when the first clock signal is input to the voltage booster, the control circuit is configured to: [0181] during a first phase of a period of the first clock signal, turn on the first switch, turn off the second switch, and couple the second terminal of the capacitor to a ground; and [0182] during a second phase of the period of the first clock signal, couple a voltage at the output of the amplifier to the second terminal of the capacitor, turn off the first switch, and turn on the second switch.

[0183] 13. The voltage regulator of clause 12, wherein, when the second clock signal is input to the voltage booster, the control circuit is configured to: [0184] during a first phase of a period of the second clock signal, turn on the first switch, turn off the second switch, and couple the second terminal of the capacitor to the ground; and [0185] during a second phase of the period of the second clock signal, couple the voltage at the output of the amplifier to the second terminal of the capacitor, turn off the first switch, and turn on the second switch.

[0186] 14. The voltage regulator of any one of clauses 1 to 13, wherein the pass transistor comprises an n-type field effect transistor (NFET).

[0187] 15. A method of operating a voltage regulator, wherein the voltage regulator includes a pass transistor, an amplifier, and a voltage booster coupled between an output of the amplifier and a gate of the pass transistor, wherein a first input of the amplifier receives a reference voltage and a second input of the amplifier receives a feedback voltage via a feedback path coupled to a source of the pass transistor, the method comprising: [0188] detecting a difference between the reference voltage and the feedback voltage is greater than a threshold voltage; and [0189] switching a clock input of the voltage booster from a first clock signal to a second clock signal in response to detecting the difference between the reference voltage and the feedback voltage is greater than the threshold voltage, wherein the second clock signal has a higher clock frequency than the first clock signal.

[0190] 16. The method of clause 15, further comprising: [0191] detecting the difference between the reference voltage and the feedback voltage is less than the threshold voltage; and [0192] switching the clock input of the voltage booster from the second clock signal to the first clock signal in response to detecting the difference between the reference voltage and the feedback voltage is less than the threshold voltage.

[0193] 17. A voltage regulator, comprising: [0194] a pass transistor coupled between an input of the

voltage regulator and an output of the voltage regulator; [0195] an amplifier having a first input, a second input, and an output, wherein the first input of the amplifier is configured to receive a reference voltage, and the second input of the amplifier is coupled to the output of the voltage regulator via a feedback path; [0196] a voltage booster coupled between the output of the amplifier and a gate of the pass transistor, wherein the voltage booster includes a boost capacitor and an output capacitor, the output capacitor is coupled to the gate of the pass transistor, and the voltage booster is configured to: [0197] during a first phase, charge the boost capacitor with current from the output of the amplifier; and [0198] during a second phase, boost a voltage on the boost capacitor and transfer charge from the boost capacitor to the output capacitor; [0199] a capacitance controller configured to switch the boost capacitor between a first capacitance and a second capacitance higher than the first capacitance; and [0200] a detection circuit having an input and an output, wherein the input of the detection circuit is coupled to the amplifier, and the output of the detection circuit is coupled to the capacitance controller.

[0201] 18. The voltage regulator of clause 17, wherein the detection circuit is configured to: [0202] detect a difference between the reference voltage at the first input of the amplifier and a feedback voltage at the second input of the amplifier; [0203] cause the capacitance controller to switch the boost capacitor to the second capacitance if the difference between the reference voltage and the feedback voltage is greater than a threshold voltage; and [0204] cause the capacitance controller to switch the boost capacitor to the first capacitance if the difference between the reference voltage and the feedback voltage is less than the threshold voltage.

[0205] 19. The voltage regulator of clause 18, wherein the input of the detection circuit includes a first input coupled to the first input of the amplifier and a second input coupled to the second input of the amplifier.

[0206] 20. The voltage regulator of clause 17, wherein: [0207] the amplifier comprises an input stage configured to: [0208] convert the reference voltage at the first input of the amplifier into a first current; and [0209] convert a feedback voltage at the second input of the amplifier into a second current; and [0210] the detection circuit comprises a current mirror configured to: [0211] generate a pull-up current at the output of the detection circuit that is proportional to the first current by a first scaling factor; and [0212] generate a pull-down current at the output of the detection circuit that is proportional to the second current by a second scaling factor greater than the first scaling factor.

[0213] 21. The voltage regulator of clause 20, wherein the capacitance controller is configured to: [0214] switch the boost capacitor to the second capacitance if the output of the detection circuit is logic one; and [0215] switch the boost capacitor to the first capacitance if the output of the detection circuit is logic zero.

[0216] 22. The voltage regulator of any one of clauses 17 to 21, wherein: [0217] the boost capacitor includes a first capacitor and a second capacitor; and [0218] the capacitance controller comprises a switch coupled in series with the second capacitor.

[0219] 23. The voltage regulator of clause 22, wherein the first capacitance is equal to a capacitance of the first capacitor, and the second capacitance is equal to a sum of the capacitance of the first capacitor and a capacitance of the second capacitor.

[0220] 24. The voltage regulator of clause 22 or 23, wherein the detection circuit is configured to open the switch to switch the boost capacitor to the first capacitance and close the switch to switch the boost capacitor to the second capacitance.

[0221] 25. The voltage regulator of any one of clauses 17 to 21, wherein: [0222] the boost capacitor includes a first capacitor and a second capacitor; and [0223] the capacitance controller comprises a driver coupled to the second capacitor.

[0224] 26. The voltage regulator of clause 25, wherein the detection circuit is configured to disable the driver to switch the boost capacitor to the first capacitance and enable the driver to switch the boost capacitor to the second capacitance.

[0225] 27. The voltage regulator of clause 25 or 26, wherein the driver comprises a tri-state inverter.

[0226] 28. The voltage regulator of any one of clauses 17 to 27, further comprising a pulse stretcher coupled between the detection circuit and the capacitance controller.

[0227] 29. The voltage regulator of any one of clauses 17 to 28, wherein the pass transistor comprises an n-type field effect transistor (NFET).

[0228] 30. A method of operating a voltage regulator, wherein the voltage regulator includes a pass transistor, an amplifier, and a voltage booster coupled between an output of the amplifier and a gate of the pass transistor, wherein a first input of the amplifier receives a reference voltage and a second input of the amplifier receives a feedback voltage via a feedback path coupled to a source of the pass transistor, the method comprising: [0229] detecting a difference between the reference voltage and the feedback voltage is greater than a threshold voltage; and [0230] switching a boost capacitor of the voltage booster from a first capacitance to a second capacitance in response to detecting the difference between the reference voltage and the feedback voltage is greater than the threshold voltage, wherein the second capacitance is higher than the first capacitance.

[0231] 31. The method of clause 30, further comprising: [0232] detecting the difference between the reference voltage and the feedback voltage is less than the threshold voltage; and [0233] switching the boost capacitor of the voltage booster from the second capacitance to the first capacitance in response to detecting the difference between the reference voltage and the feedback voltage is less than the threshold voltage.

[0234] Any reference to an element herein using a designation such as “first,” “second,” and so forth does not generally limit the quantity or order of those elements. Rather, these designations are used herein as a convenient way of distinguishing between two or more elements or instances of an element. Thus, a reference to first and second elements does not mean that only two elements can be employed, or that the first element must precede the second element.

[0235] Within the present disclosure, the word “exemplary” is used to mean “serving as an example, instance, or illustration.” Any implementation or aspect described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects of the disclosure. Likewise, the term “aspects” does not require that all aspects of the disclosure include the discussed feature, advantage or mode of operation. The term “approximately”, as used herein with respect to a stated value or a property, is intended to indicate being within 10% of the stated value or property (i.e., between 90% to 110% of the stated value or property).

[0236] The previous description of the disclosure is provided to enable any person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations without departing from the spirit or scope of the disclosure. Thus, the disclosure is not intended to be limited to the examples described herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

Claims

1. A voltage regulator, comprising: a pass transistor coupled between an input of the voltage regulator and an output of the voltage regulator; an amplifier having a first input, a second input, and an output, wherein the first input of the amplifier is configured to receive a reference voltage, and the second input of the amplifier is coupled to the output of the voltage regulator via a feedback path; a voltage booster coupled between the output of the amplifier and a gate of the pass transistor; a multiplexer having a first input, a second input, an output, and a select input, wherein the first input of the multiplexer is configured to receive a first clock signal, the second input of the multiplexer is configured to receive a second clock signal having a higher frequency than the first clock signal, and the output of the multiplexer is coupled to a clock input of the voltage booster;

and a detection circuit having an input and an output, wherein the input of the detection circuit is coupled to the amplifier, and the output of the detection circuit is coupled to the select input of the multiplexer.

2. The voltage regulator of claim 1, wherein the voltage booster comprises a charge pump.

3. The voltage regulator of claim 1, wherein the detection circuit is configured to: detect a difference between the reference voltage at the first input of the amplifier and a feedback voltage at the second input of the amplifier; cause the multiplexer to select the second clock signal if the difference between the reference voltage and the feedback voltage is greater than a threshold voltage; and cause the multiplexer to select the first clock signal if the difference between the reference voltage and the feedback voltage is less than the threshold voltage.

4. The voltage regulator of claim 1, wherein: the amplifier comprises an input stage configured to: convert the reference voltage at the first input of the amplifier into a first current; and convert a feedback voltage at the second input of the amplifier into a second current; and the detection circuit comprises a current mirror configured to: generate a pull-up current at the output of the detection circuit that is proportional to the first current by a first scaling factor; and generate a pull-down current at the output of the detection circuit that is proportional to the second current by a second scaling factor greater than the first scaling factor.

5. The voltage regulator of claim 4, wherein the multiplexer is configured to: select the second clock signal if the output of the detection circuit is logic one; and select the first clock signal if the output of the detection circuit is logic zero.

6. The voltage regulator of claim 1, wherein the voltage booster comprises: a capacitor having a first terminal and a second terminal; a first switch coupled between the output of the amplifier and the first terminal of the capacitor; a second switch coupled between the first terminal of the capacitor and the gate of the pass transistor; and a control circuit coupled to the clock input of the voltage booster, a control input of the first switch, a control input of the second switch, and the second terminal of the capacitor.

7. The voltage regulator of claim 6, wherein, when the first clock signal is input to the voltage booster, the control circuit is configured to: during a first phase of a period of the first clock signal, turn on the first switch, turn off the second switch, and couple the second terminal of the capacitor to a ground; and during a second phase of the period of the first clock signal, couple a voltage at the output of the amplifier to the second terminal of the capacitor, turn off the first switch, and turn on the second switch.

8. The voltage regulator of claim 7, wherein, when the second clock signal is input to the voltage booster, the control circuit is configured to: during a first phase of a period of the second clock signal, turn on the first switch, turn off the second switch, and couple the second terminal of the capacitor to the ground; and during a second phase of the period of the second clock signal, couple the voltage at the output of the amplifier to the second terminal of the capacitor, turn off the first switch, and turn on the second switch.

9. A voltage regulator, comprising: a pass transistor coupled between an input of the voltage regulator and an output of the voltage regulator; an amplifier having a first input, a second input, and an output, wherein the first input of the amplifier is configured to receive a reference voltage, and the second input of the amplifier is coupled to the output of the voltage regulator via a feedback path; a voltage booster coupled between the output of the amplifier and a gate of the pass transistor, wherein the voltage booster includes a boost capacitor and an output capacitor, the output capacitor is coupled to the gate of the pass transistor, and the voltage booster is configured to: during a first phase, charge the boost capacitor with current from the output of the amplifier; and during a second phase, boost a voltage on the boost capacitor and transfer charge from the boost capacitor to the output capacitor; a capacitance controller configured to switch the boost capacitor between a first capacitance and a second capacitance higher than the first capacitance; and a detection circuit having an input and an output, wherein the input of the detection circuit is

coupled to the amplifier, and the output of the detection circuit is coupled to the capacitance controller.

10. The voltage regulator of claim 9, wherein the detection circuit is configured to: detect a difference between the reference voltage at the first input of the amplifier and a feedback voltage at the second input of the amplifier; cause the capacitance controller to switch the boost capacitor to the second capacitance if the difference between the reference voltage and the feedback voltage is greater than a threshold voltage; and cause the capacitance controller to switch the boost capacitor to the first capacitance if the difference between the reference voltage and the feedback voltage is less than the threshold voltage.

11. The voltage regulator of claim 9, wherein: the amplifier comprises an input stage configured to: convert the reference voltage at the first input of the amplifier into a first current; and convert a feedback voltage at the second input of the amplifier into a second current; and the detection circuit comprises a current mirror configured to: generate a pull-up current at the output of the detection circuit that is proportional to the first current by a first scaling factor; and generate a pull-down current at the output of the detection circuit that is proportional to the second current by a second scaling factor greater than the first scaling factor.

12. The voltage regulator of claim 11, wherein the capacitance controller is configured to: switch the boost capacitor to the second capacitance if the output of the detection circuit is logic one; and switch the boost capacitor to the first capacitance if the output of the detection circuit is logic zero.

13. The voltage regulator of claim 9, wherein: the boost capacitor includes a first capacitor and a second capacitor; and the capacitance controller comprises a switch coupled in series with the second capacitor.

14. The voltage regulator of claim 13, wherein the first capacitance is equal to a capacitance of the first capacitor, and the second capacitance is equal to a sum of the capacitance of the first capacitor and a capacitance of the second capacitor.

15. The voltage regulator of claim 13, wherein the detection circuit is configured to open the switch to switch the boost capacitor to the first capacitance and close the switch to switch the boost capacitor to the second capacitance.

16. The voltage regulator of claim 9, wherein: the boost capacitor includes a first capacitor and a second capacitor; and the capacitance controller comprises a driver coupled to the second capacitor.

17. The voltage regulator of claim 16, wherein the detection circuit is configured to disable the driver to switch the boost capacitor to the first capacitance and enable the driver to switch the boost capacitor to the second capacitance.

18. The voltage regulator of claim 9, wherein the pass transistor comprises an n-type field effect transistor (NFET).

19. A method of operating a voltage regulator, wherein the voltage regulator includes a pass transistor, an amplifier, and a voltage booster coupled between an output of the amplifier and a gate of the pass transistor, wherein a first input of the amplifier receives a reference voltage and a second input of the amplifier receives a feedback voltage via a feedback path coupled to a source of the pass transistor, the method comprising: detecting a difference between the reference voltage and the feedback voltage is greater than a threshold voltage; and switching a boost capacitor of the voltage booster from a first capacitance to a second capacitance in response to detecting the difference between the reference voltage and the feedback voltage is greater than the threshold voltage, wherein the second capacitance is higher than the first capacitance.

20. The method of claim 19, further comprising: detecting the difference between the reference voltage and the feedback voltage is less than the threshold voltage; and switching the boost capacitor of the voltage booster from the second capacitance to the first capacitance in response to detecting the difference between the reference voltage and the feedback voltage is less than the threshold voltage.
